UNC | McALLISTER HEART INSTITUTE

Building a Functional Heart

From Cellular Neighborhoods to Gene Function

Coordinating postnatal muscle, vascular and matrix development

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New muscle must become functional myocardium

Produce or preserve muscle

Cell number and growth

Replenish the tissue that has been lost

Build durable function

Contract, conduct and sustain workload

Integrate muscle with vessels and matrix

Postnatal development reveals how growing muscle acquires function within a tissue.

Tzahor & Poss, Science 356, 1035–1039 (2017); Guo & Pu, Circ Res 126, 1086–1106 (2020)

Maturation combines several changes in cell function

Structure

Sarcomeres, cell growth and T-tubules

Electrical behavior

Excitation and calcium handling

Metabolism

Energy supply for sustained contraction

Cell-cycle behavior

One developmental feature; not a complete functional readout

Growth-associated and maturation-associated programs can overlap in the same cell.

Intrinsic programs and the local environment develop together

Intrinsic competence

Chromatin and transcriptional programs

Prior lab example:

CARM1-dependent regulation

Local tissue context

Capillaries and mural cells

Fibroblast matrix

Endocardial environment

Mechanical and paracrine cues

Changing physiology

Circulation and workload

Nutrition and hormones

Oxygen demand and supply

Shared developmental inputs

Map the tissue, nominate regulators, then test function

01 MAP

Who is present?

Where are they?

How do states vary?

→

02 CONNECT

Which vascular and matrix

contexts accompany

CM maturation?

→

03 TEST

Which candidate genes

are required for

cell and heart function?

An atlas connects four developmental stages across two assays

P0 **P7** **P14** **P21**

Single-nucleus RNA-seq

Broad transcriptomic coverage

2 libraries / 2 animals per age

Xenium spatial transcriptomics

403 measured gene targets

1 observed section per age

RNA supports cell identities; measured Xenium counts support spatial comparisons.

Keep measured expression separate from transferred information

Directly measured

Xenium target counts

The 21-gene spatial maturation module

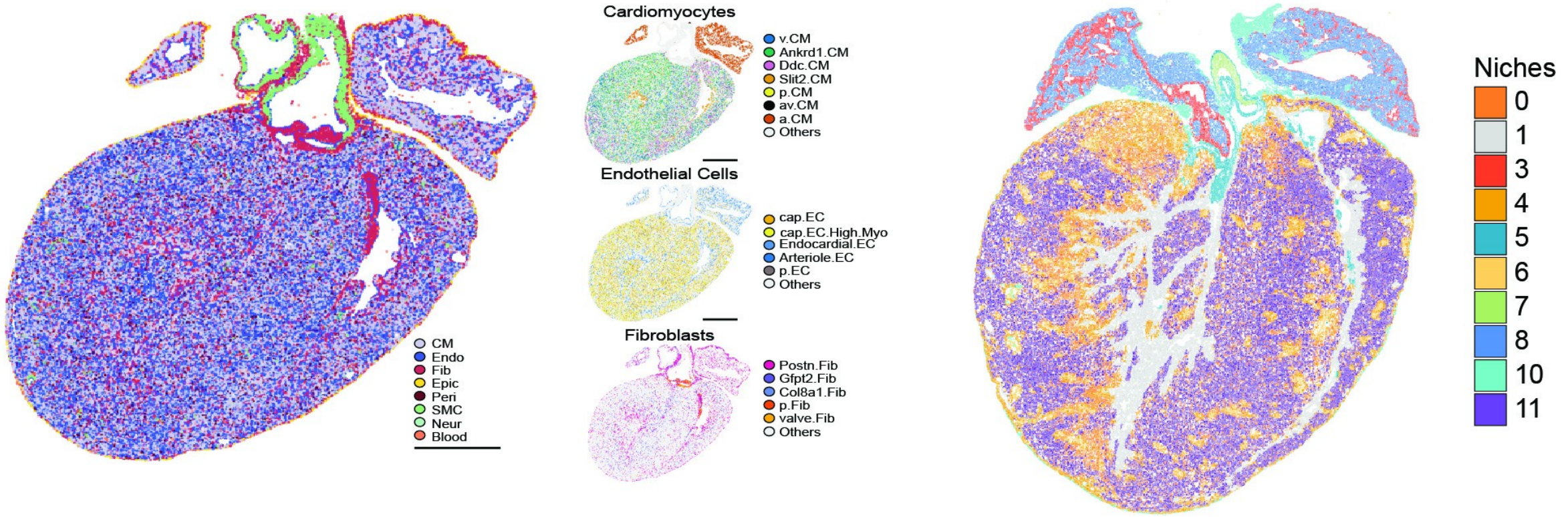
Reference-supported / inferred

Cell identities and subtypes

Broader RNA programs placed into spatial context

No missing maturation genes are imputed into the spatial 21-gene outcome.

Cardiac cells occupy organized tissue neighborhoods



Organized neighborhoods provide biological context and a potential source of regional confounding.

Subtypes make the neighborhood biologically informative

Cardiomyocytes

Working ventricular

Ankrd1-associated

Regional / cycling

Identity ≠ maturity

Endothelial cells

Capillary-like

Arterial-like / venous-like

Endocardial-like

Lymphatic-like, if supported

Fibroblasts

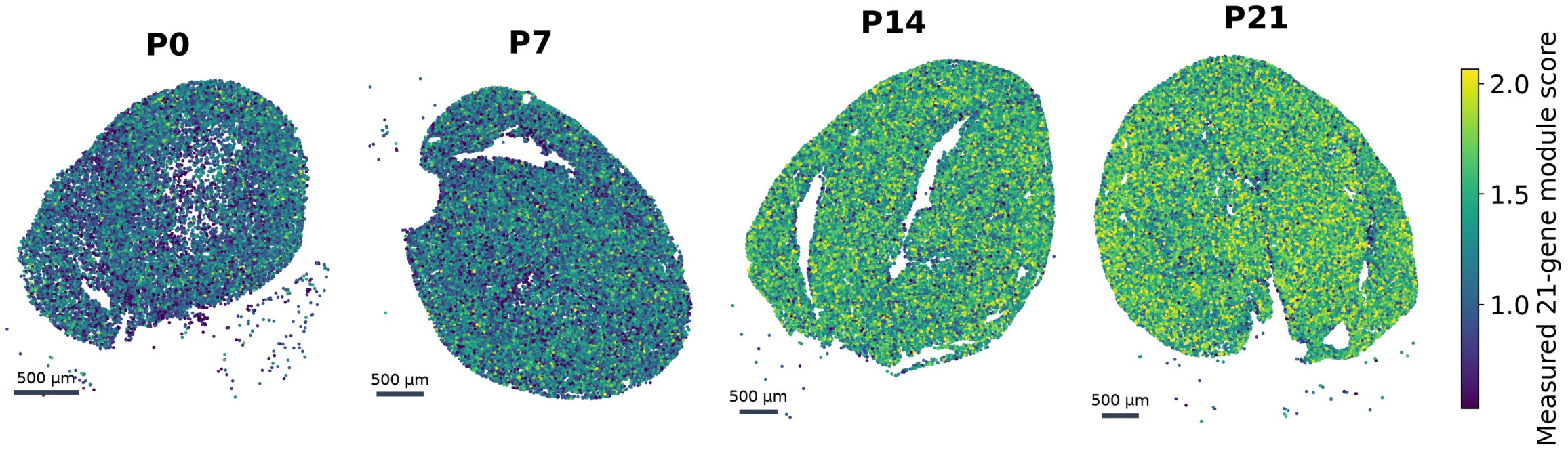
Postn-associated

Gfpt2/Pi16-associated

Col8a1-associated

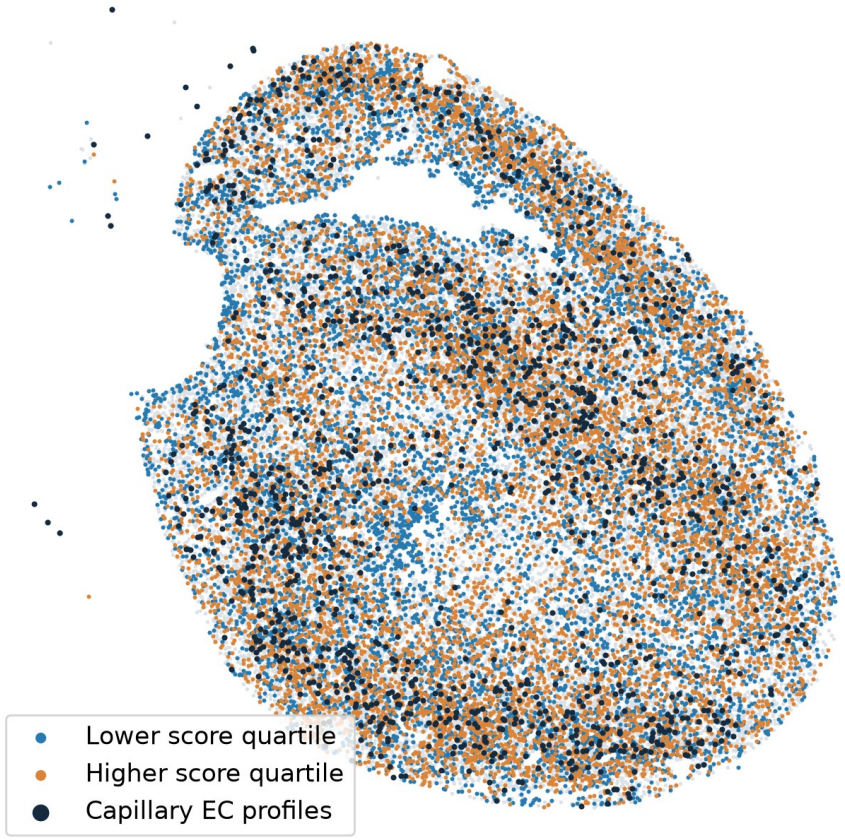
Other matrix / adhesion families

The 21-gene landscape changes with age



The same measured 21-gene module is used across ages; the shared color scale shows the index.

CMs in the same section have different maturation scores

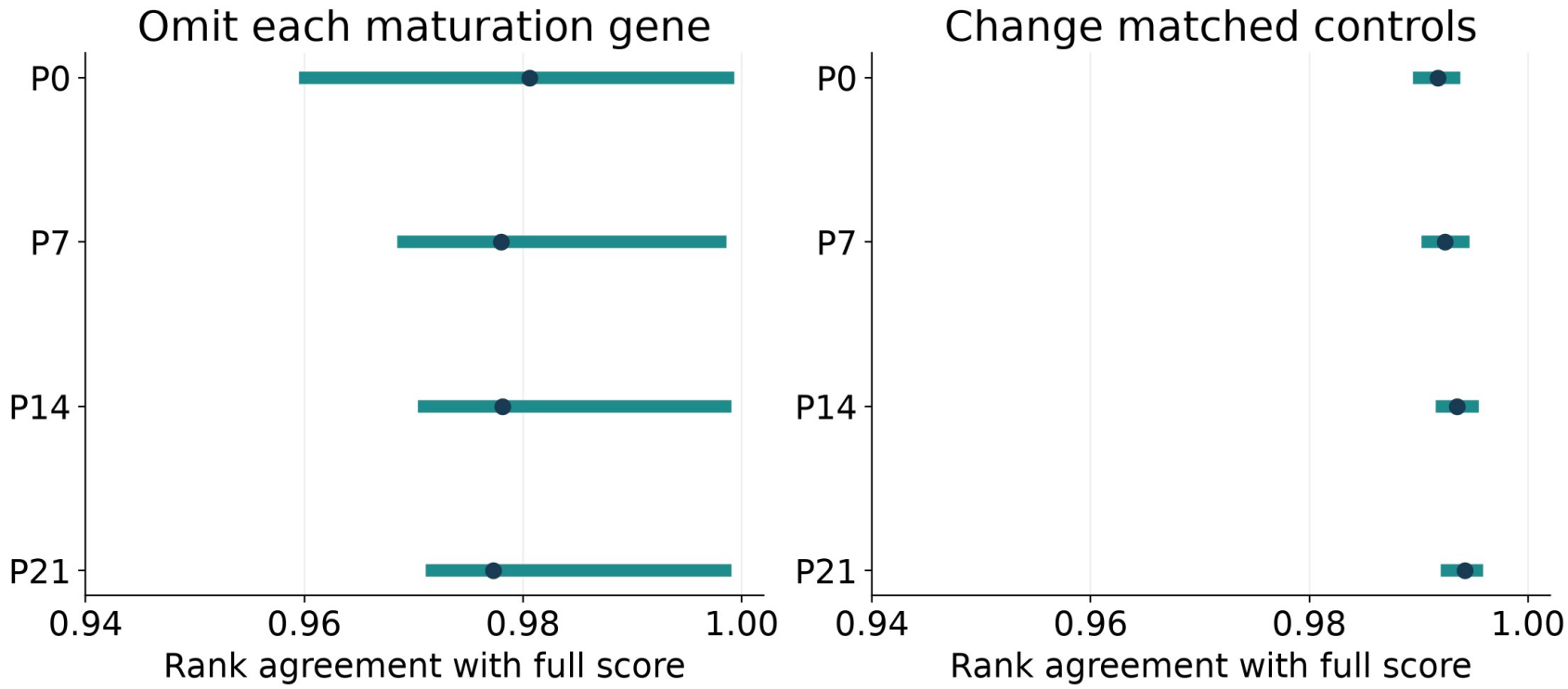


Variation within one age

P7 lower and higher score quartiles

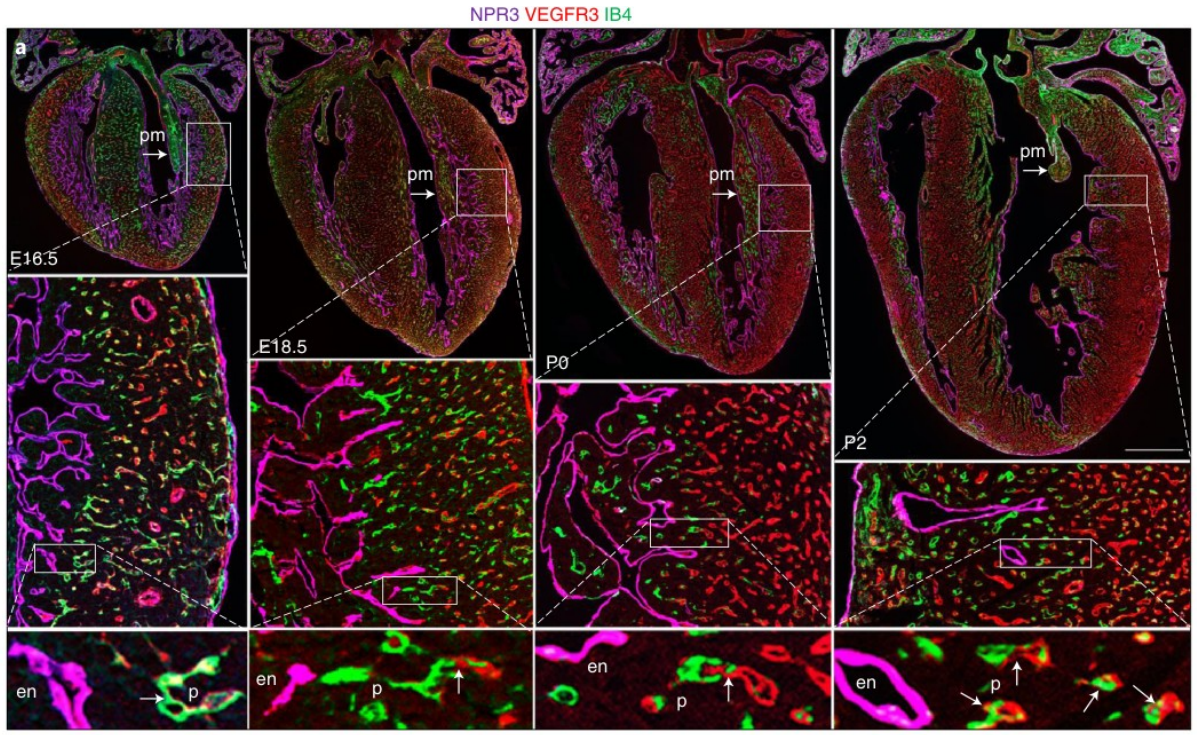
Relative transcriptional states within the same section

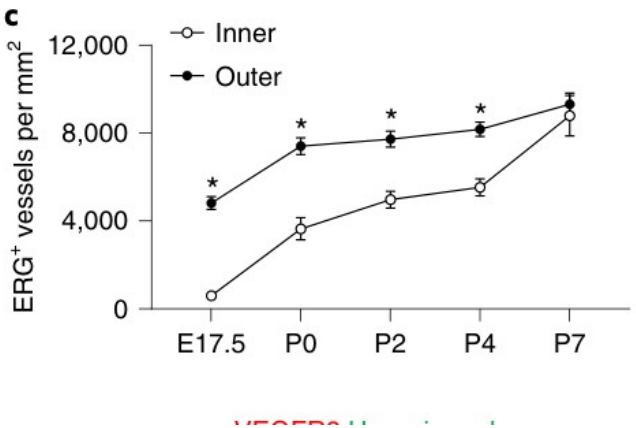
The score is stable to individual-gene omission



Stable to individual genes and control draws; transcript ownership and source labels remain uncertain.

Coronary growth is regional and begins before birth





Inner-wall vessel density approaches outer-wall density by P7

Pre-existing coronary vessels expand into growing inner myocardium; the process begins before birth.

Vascular expansion and CM maturation can overlap

Late fetal → P0

Regional coronary ingrowth is underway

P0 → P7

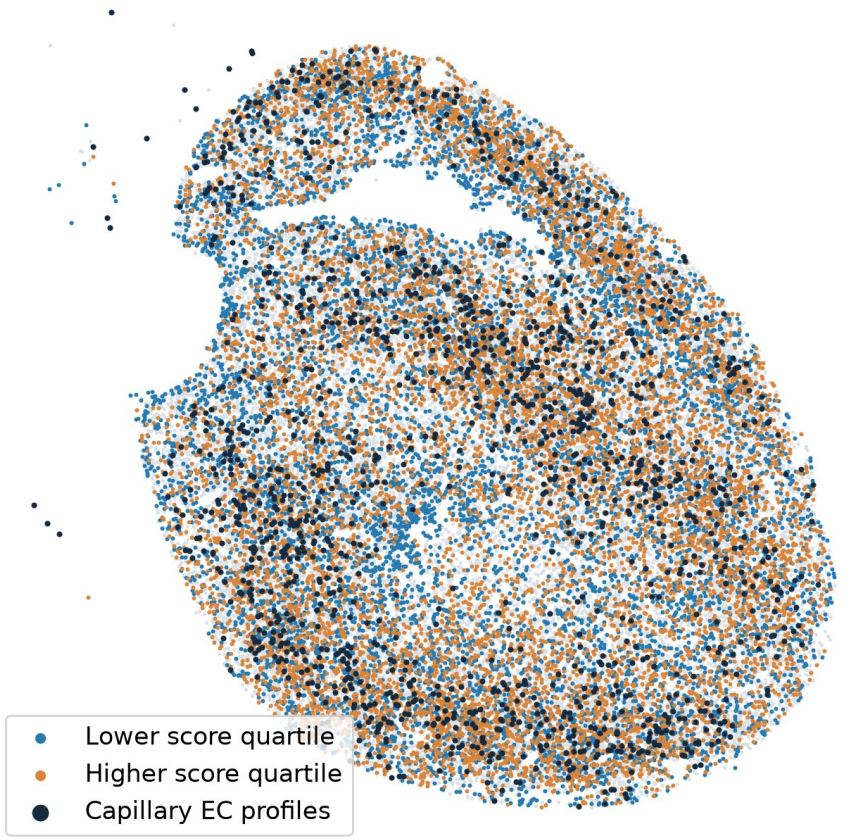
Vascular expansion and muscle growth overlap

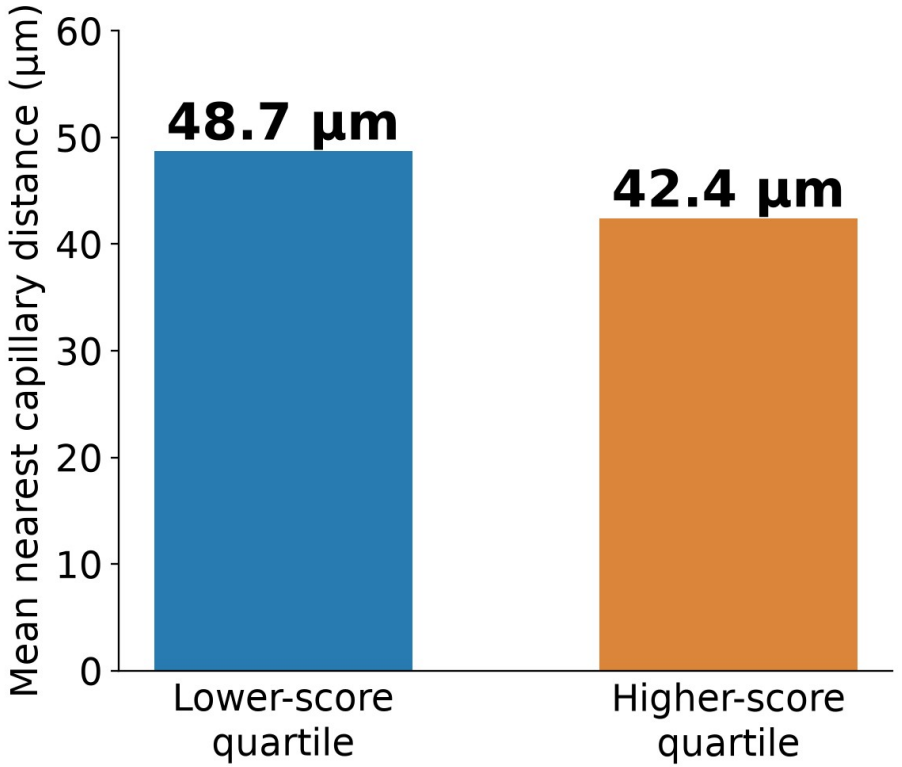
P7 → P21

CM specialization continues as vascular context changes

Schematic developmental framework: vessel growth is not a direct measurement of vascular delivery.

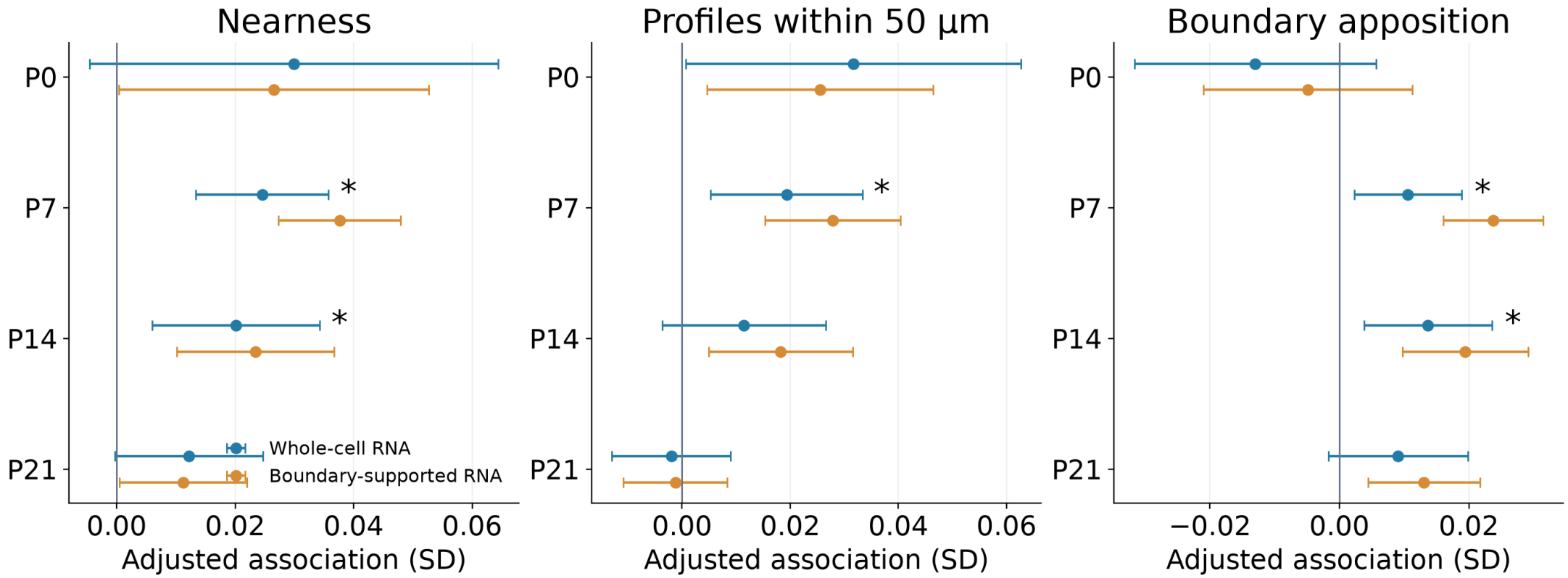
At P7, higher-scoring CMs are closer to capillary ECs





P7: positive individual-CM and locally adjusted associations; small effects in one observed section.

Nearness and boundary apposition give complementary evidence



P7: all three metrics pass correction. P14: nearness and apposition pass; annotation sensitivity remains.

Local comparisons reduce a shared-anatomy explanation

Comparable CMs

Same stage

Same available CM subtype

Comparable local tissue

Within nearby regions

No anatomical exclusion

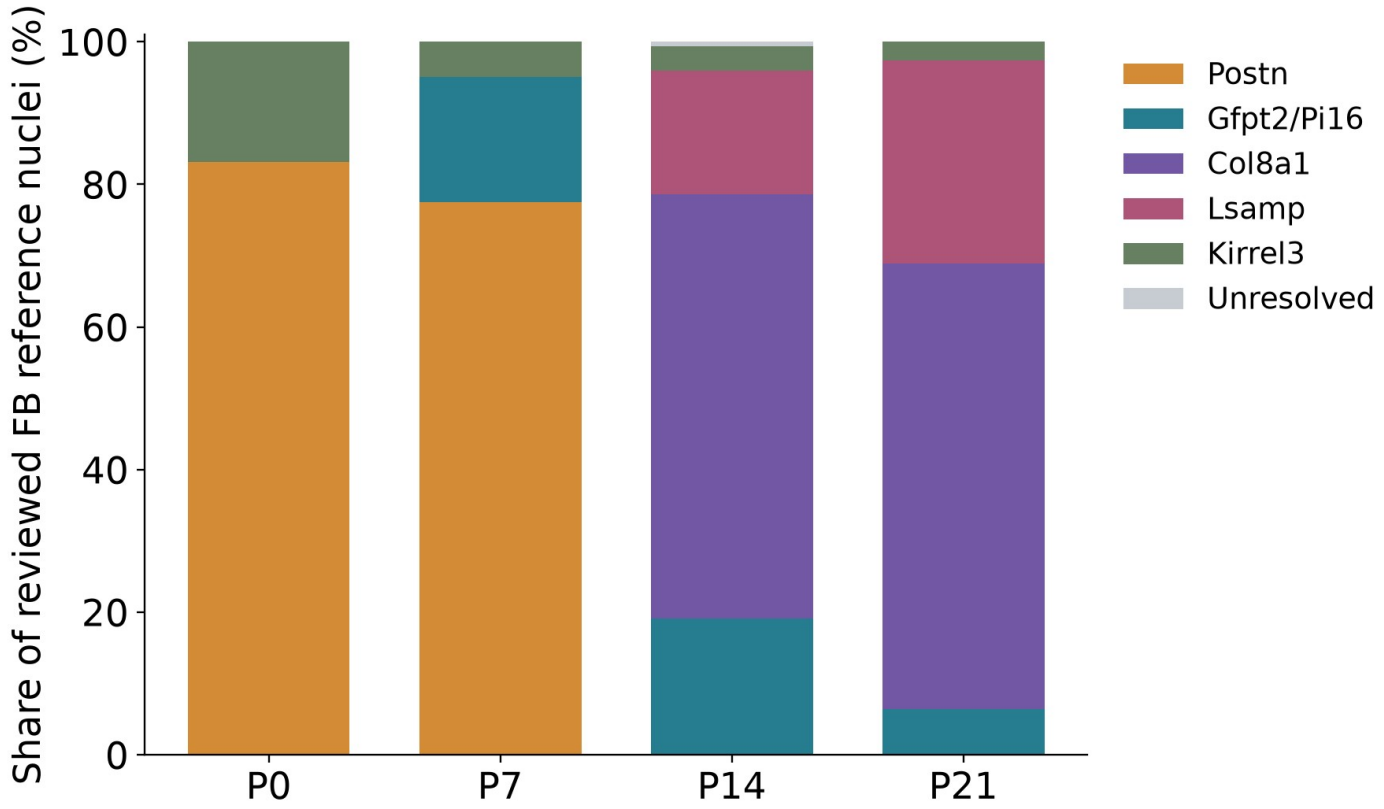
Technical context

RNA capture and cell size

Nucleus-free CMs retained

These comparisons reduce regional confounding; exact author anatomical masks remain unavailable.

Fibroblast families change as the ventricle develops

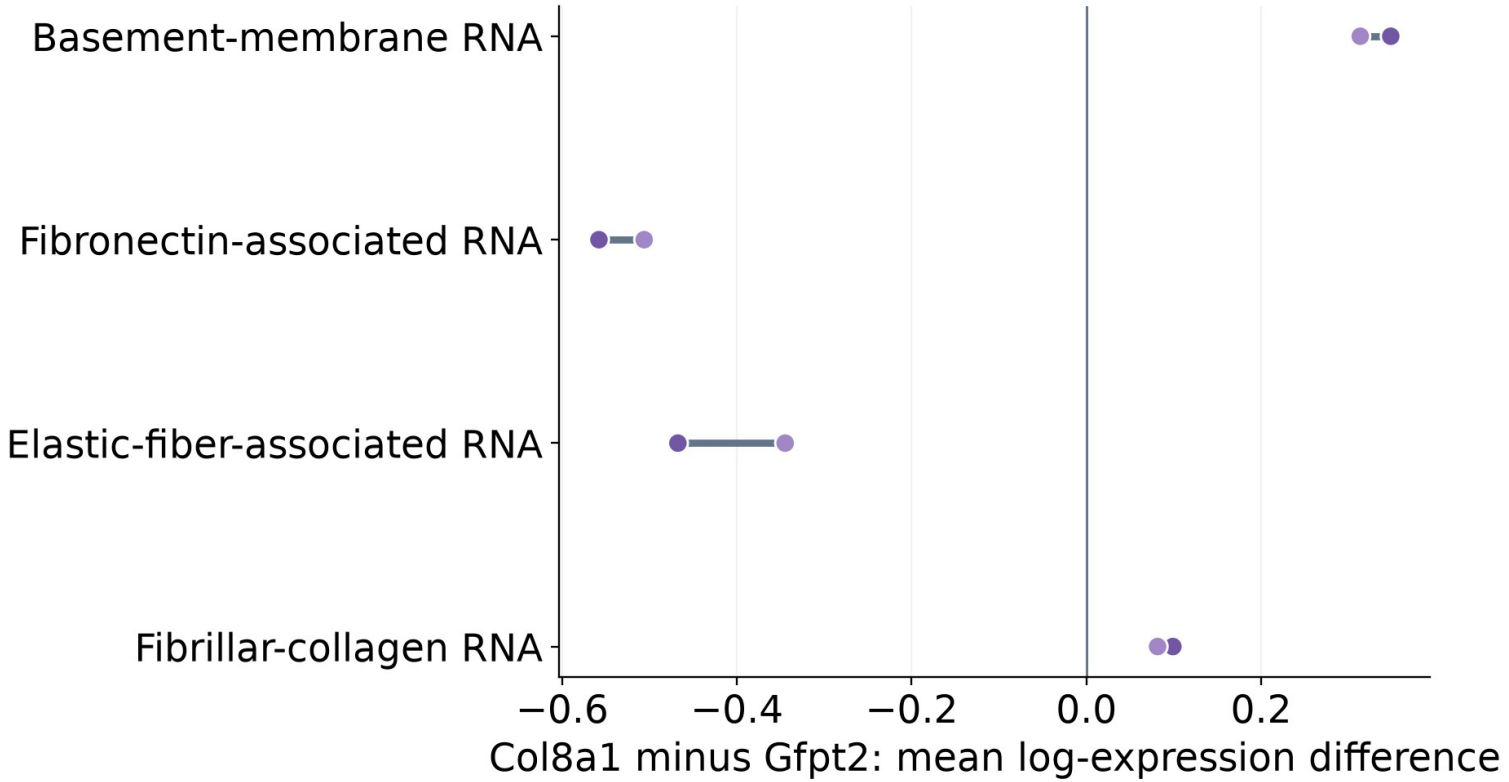


Early developmental matrix programs

Later matrix-associated specialization

Reference-nucleus shares are descriptive; changing family labels do not establish cell fate or disappearance.

Late fibroblast families emphasize different matrix programs



Col8a1 family

Basement-membrane RNA relatively higher

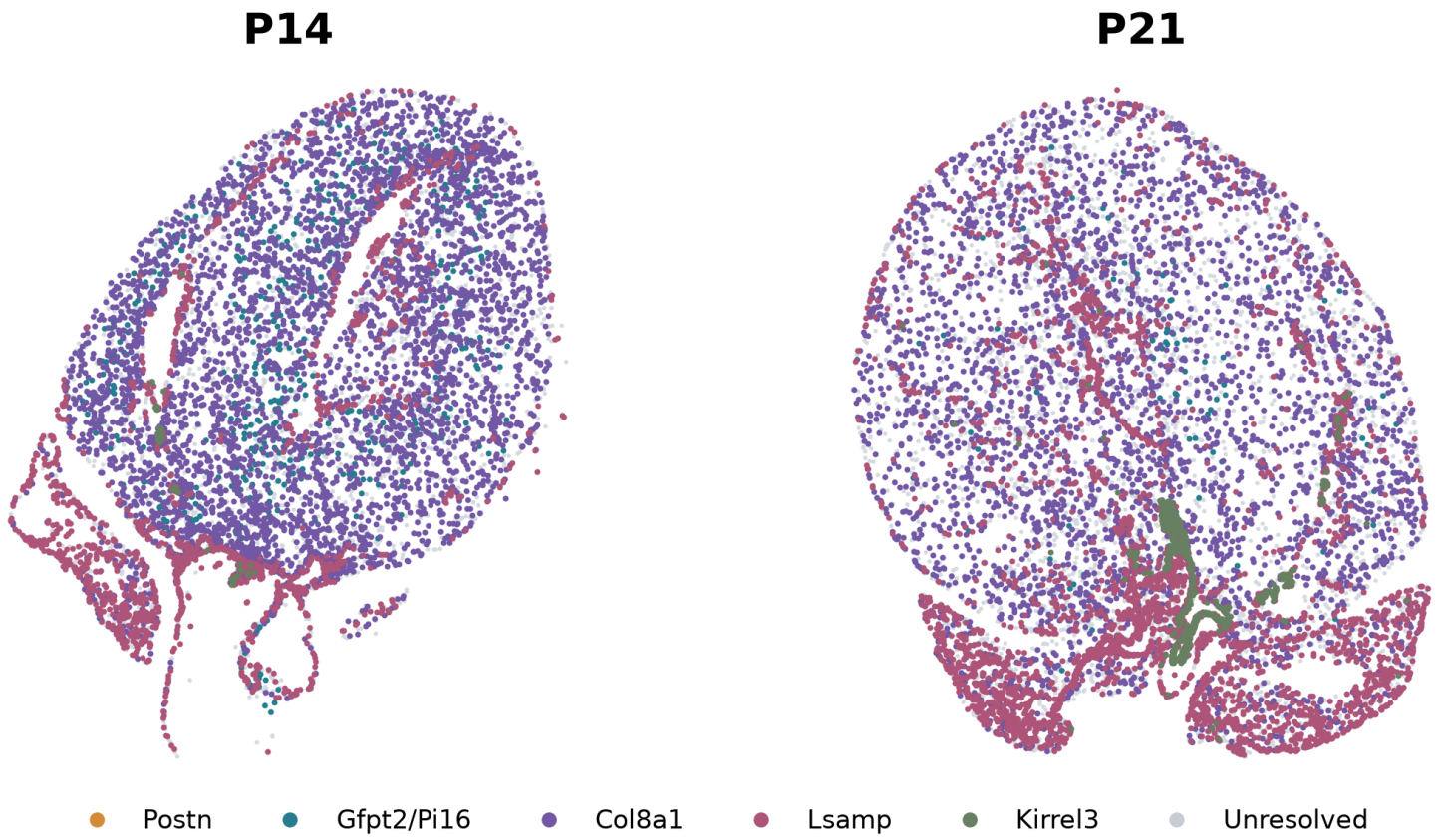
Gfpt2/Pi16 family

Fibronectin / elastic programs relatively higher

P14 paired libraries: RNA specialization suggests matrix roles; deposited matrix and mechanics are unmeasured.

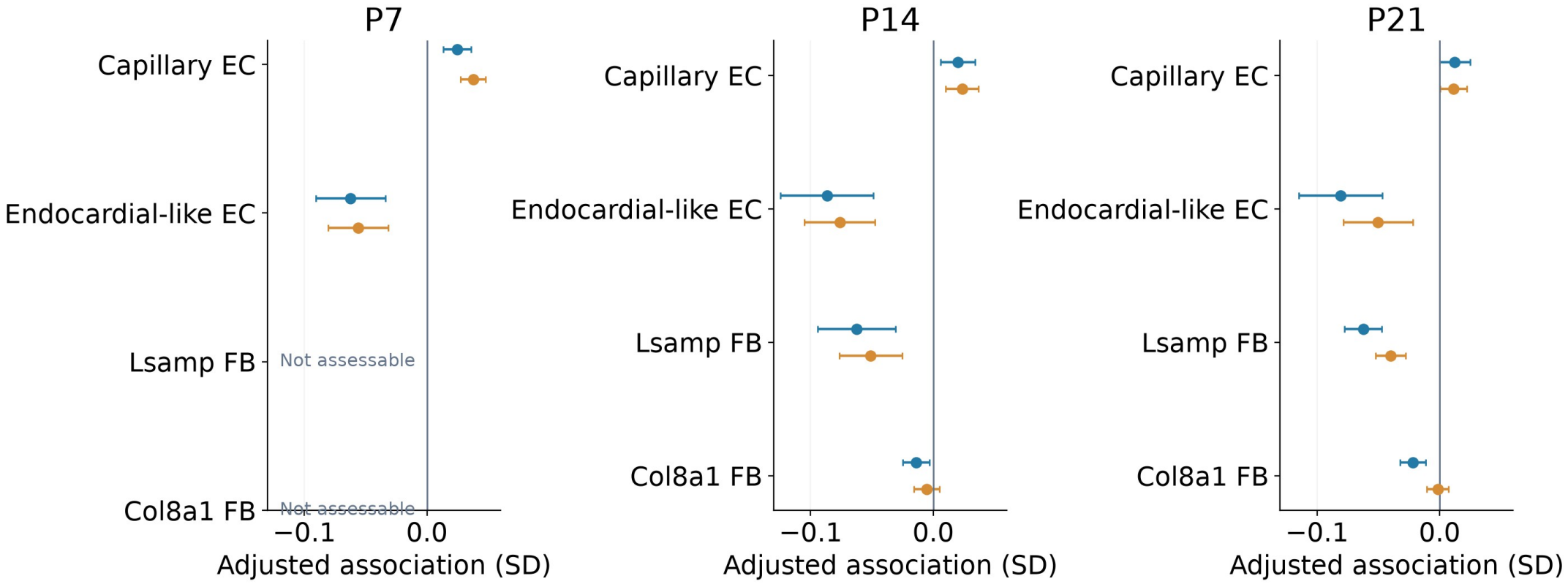
fibroblast_states/RNA_matrix_direct_family_pairs.tsv; P14 corrected comparisons

Fibroblast families occupy different spatial territories



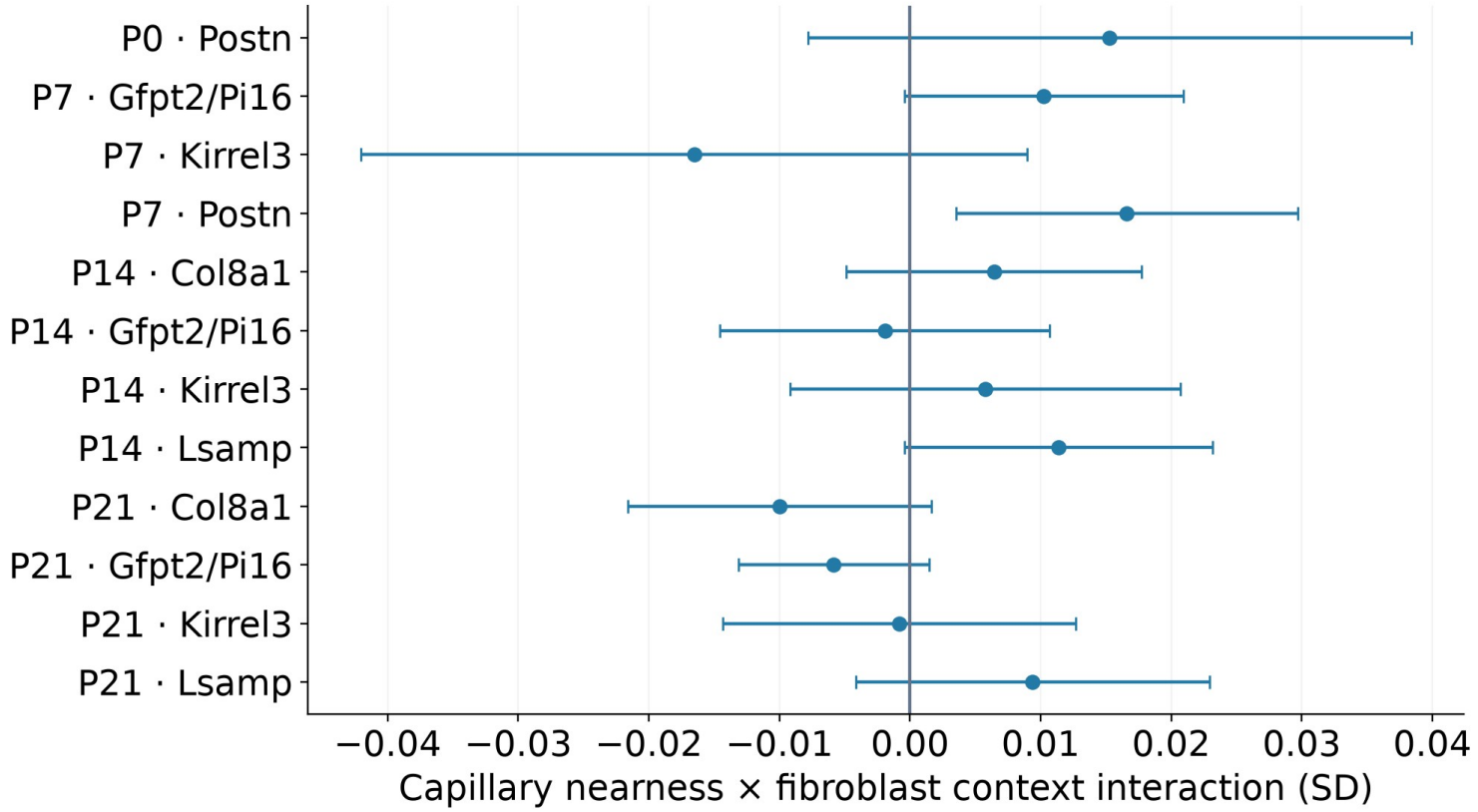
Candidate labels; unresolved profiles remain visible. P21 Gfpt2 placement is particularly incomplete.

The maturation relationship depends on the neighboring subtype



Whole-cell (blue) and boundary-supported RNA (gold): associations do not establish promotion or inhibition.

A vascular–matrix interaction remains a hypothesis to test



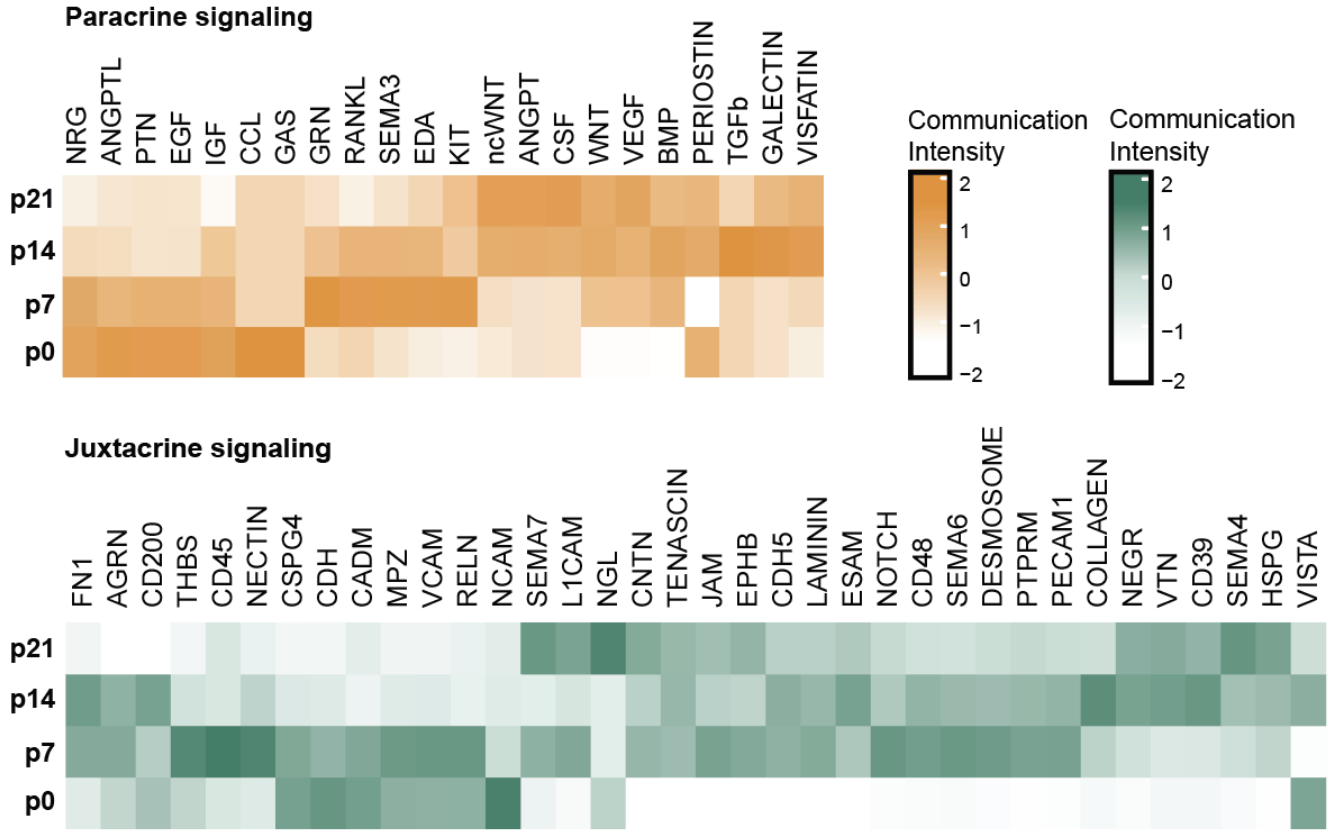
No corrected interaction

12 assessable tests

A combined vascular–matrix

effect remains unresolved

Spatial observations nominate signals for functional testing



What the model asks

Who can send?

Who can receive?

Are they nearby?

Which CM response covaries with the signal?

Spatial and regulatory models nominate experiments; intervention tests a different level of evidence.

Wang et al., Nature Cardiovascular Research (2026); supplied slide 7; COMMOT, SCENIC and SigNET analyses

A contact-associated candidate links the niche to CM regulation

Spatial evidence

PTPRM-compatible EC–CM interfaces

Protein detected at boundaries

Perturbation evidence

Ptprm perturbation changes cell-cycle and maturation-associated RNA at P8

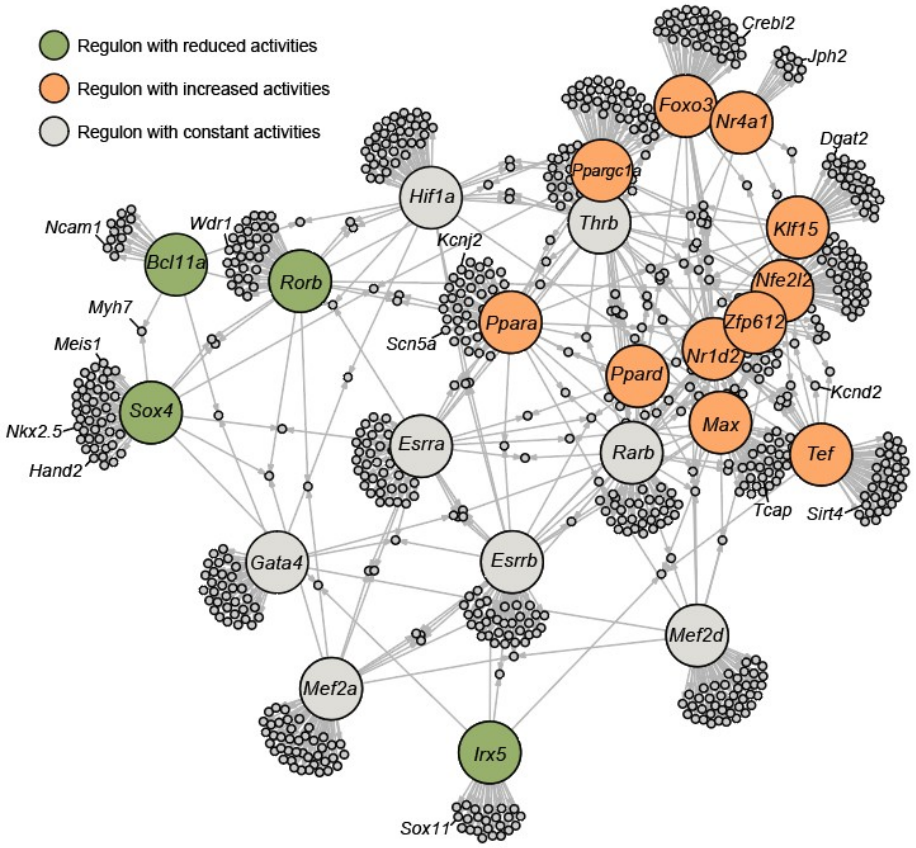
Working link

MEIS1-associated cell-cycle exit

Sender-specific mechanism and mediation remain open

A contact-associated candidate is supported; a complete EC → CM → mediator causal chain is not established.

Intrinsic regulatory networks provide another set of candidates



32

CM-specific regulons

7

Previously linked

25

Candidates

Temporal regulon inference generates a testable candidate set, not a validated binding network.

The next step is to test candidates in the intact heart

Atlas-nominated candidates

Intrinsic transcriptional regulators

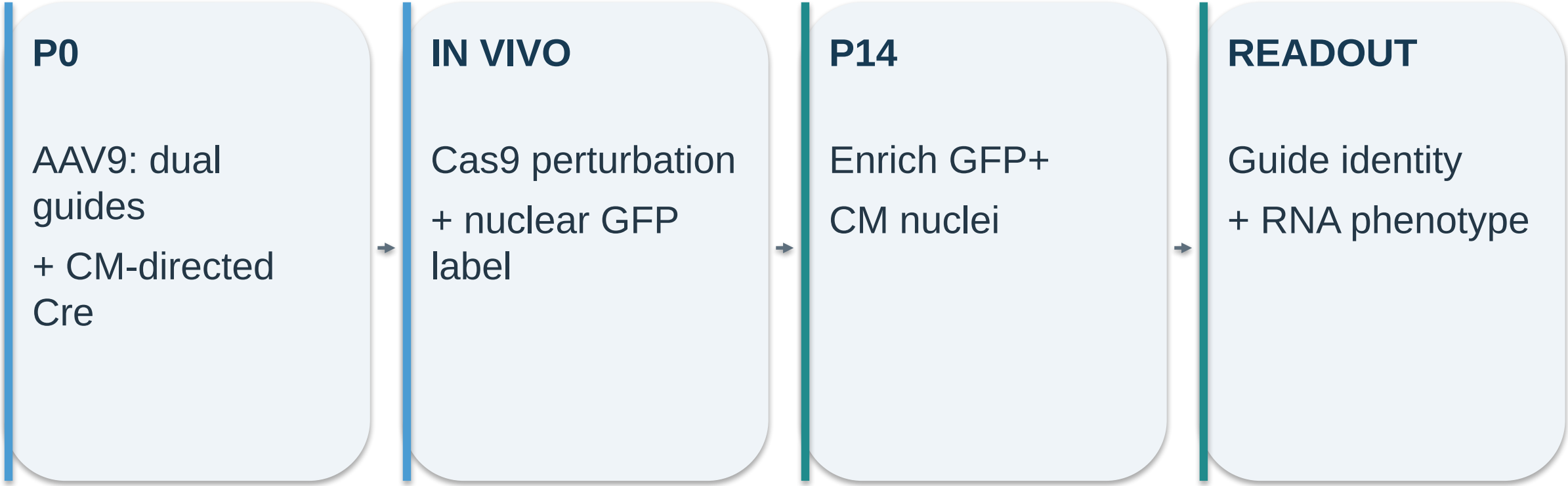
Intercellular signaling components

One scalable in vivo test

Perturb candidates in the postnatal heart

Read the RNA phenotype from identified CM nuclei

PIP-seq connects an in vivo perturbation to its RNA phenotype



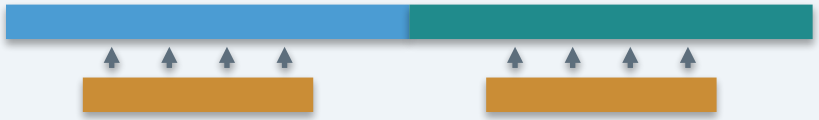
PIP-seq: probe-based, indel-detectable Perturb-seq in the native postnatal heart.

Two probe readouts strengthen perturbation assignment

Guide-detection probes

Guide transcript

Guide-specific region Constant region



Probe pairing recovers perturbation identity

Perturbation-detection probes

WT site



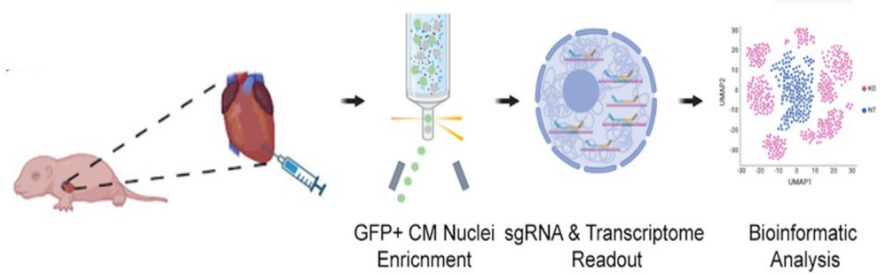
Indel

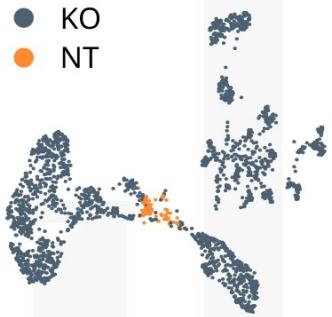


Reduced target-probe signal adds RNA-based editing evidence

Guide identity and target-probe reduction are complementary; neither alone proves biallelic editing.

The screen recovers known regulators and nominates new ones





50

Candidate regulators

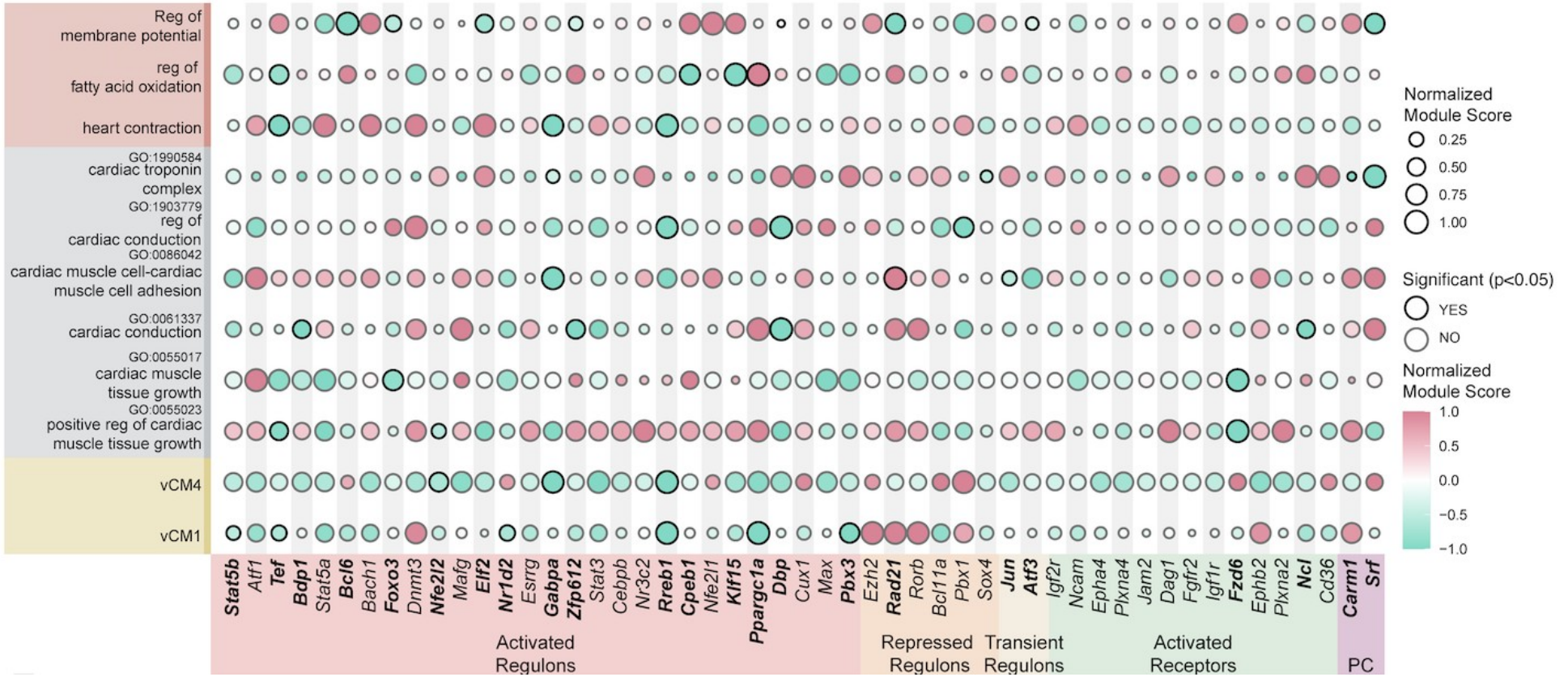
5,706

Guide-assigned nuclei

23 RNA-program hits

(2 known controls)

Different perturbations affect different maturation programs

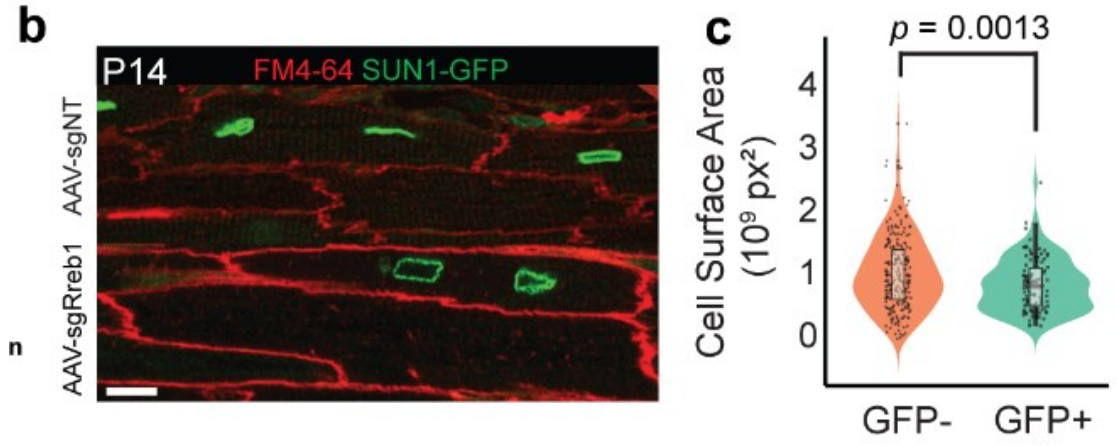


PIP-seq phenotype modules are broader RNA programs, distinct from the spatial 21-gene Xenium score.

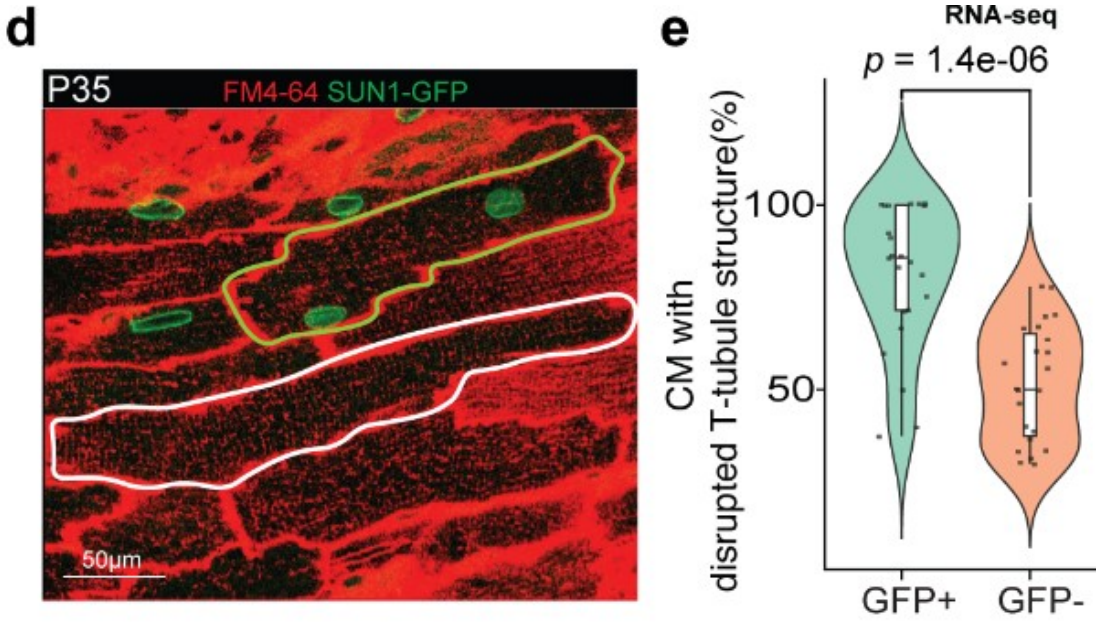
Wang et al., Nature Cardiovascular Research (2026); Fig.6e–g; original supplied slide 14

Rreb1 supports CM growth and T-tubule organization

P14 · Cardiomyocyte growth



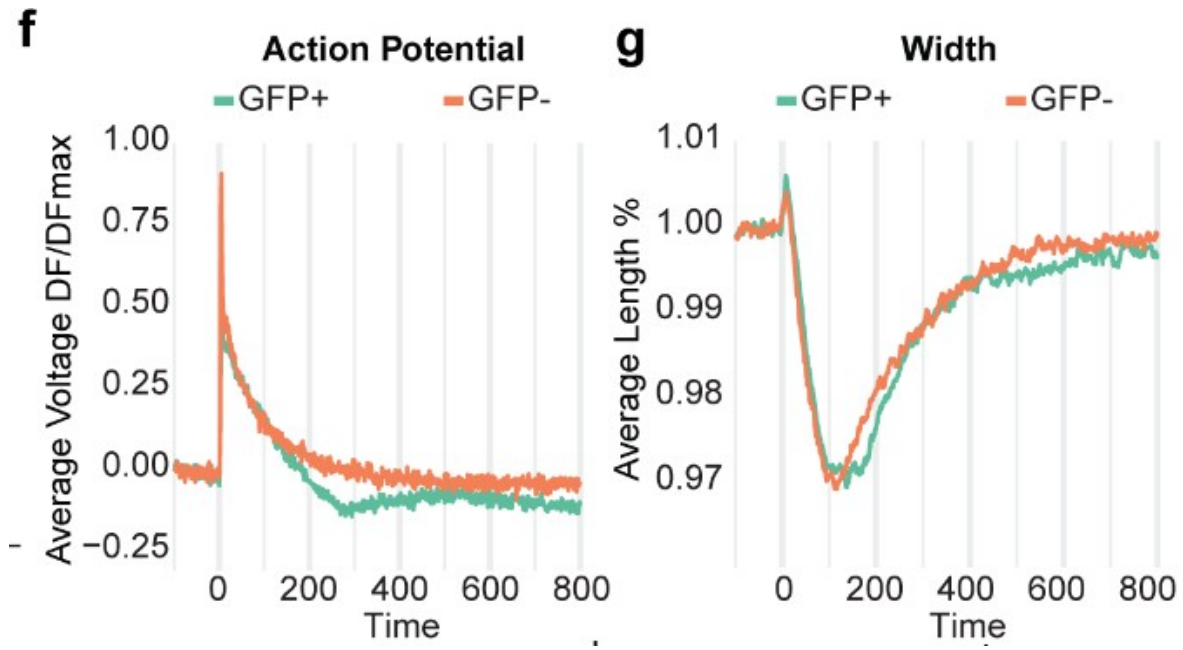
P35 · T-tubule organization



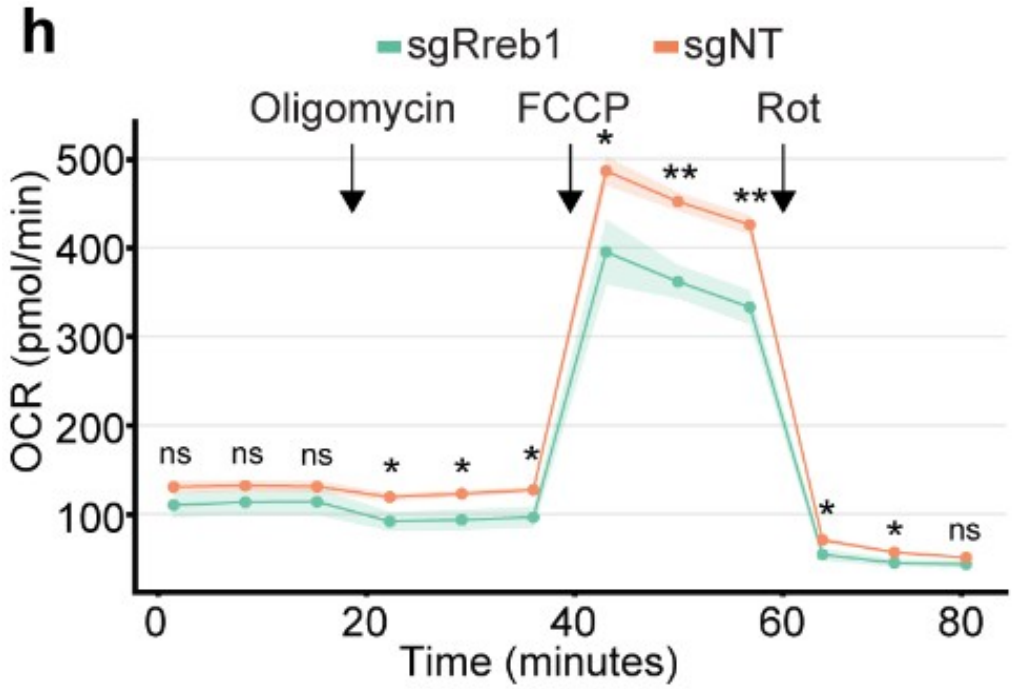
Targeted postnatal perturbation links a screen candidate to structural CM phenotypes.

Electrical and metabolic measurements extend the phenotype

Electrical and contraction readouts

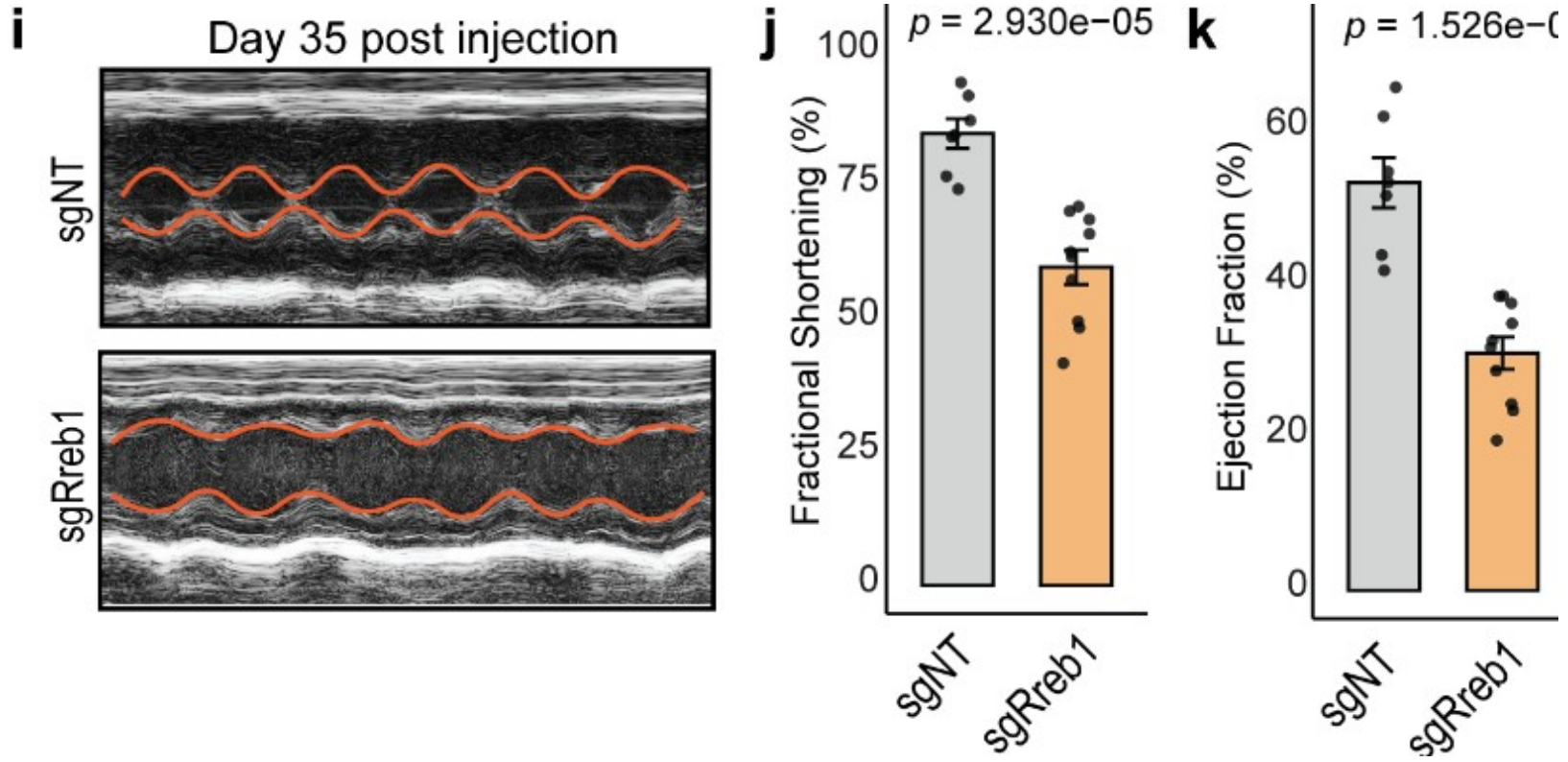


Mitochondrial respiration



Independent assay types extend the RNA-screen phenotype; fluorescence and contraction are not direct force measurements.

Rreb1 perturbation also reduces heart-level performance



Heart-level outcome

Fractional shortening ↓

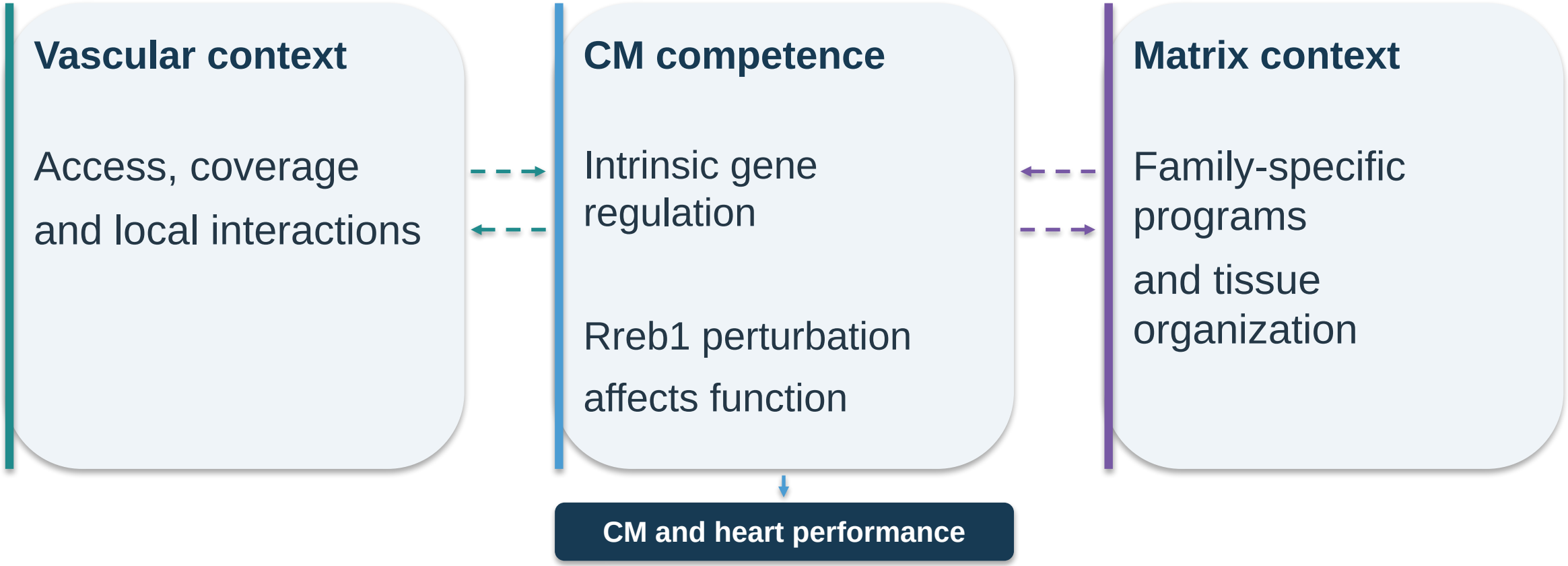
Ejection fraction ↓

7 control mice

10 sgRreb1 mice

A thyroid-hormone-associated downstream program is nominated; causal mediation still requires testing.

Mature muscle emerges within a developing tissue system



Solid: tested perturbation outcome. Dashed: proposed reciprocal niche relationships.

Development suggests goals for regenerative strategies

Generate muscle

Cell production and survival

Build the environment

Vascular access and matrix

Consolidate function

Structure, electrical behavior

and metabolic capacity

Overlapping developmental goals guide a regenerative hypothesis; no treatment efficacy is claimed.

Guo & Pu (2020); Tzahor & Poss (2017); Wang et al., Nature Cardiovascular Research (2026)

Three experiments can distinguish the remaining explanations

1 Vascular support

Independent hearts

Matched ventricular regions

Perfused-capillary measures

21-gene score + CM function

2 Contact signaling

EC- versus CM-specific perturbation

Downstream rescue

Separate source and receiver

3 Matrix organization

Defined FB state / matrix

Regional anatomy controls

Vascular and CM readouts

Intermediate P3/P5/P10 stages

A functional heart requires both a tissue map and gene tests

01

Maturation has a spatial landscape

Age and CM identity do not explain all within-section variation.

02

Neighbor identity sharpens the biological question

Capillary organization is associated; finer matrix mechanisms remain hypotheses.

03

Functional genetics moves the story forward

Targeted validation connects a candidate gene to cell and heart performance.

Acknowledgments

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- Paige Takasugi, BA
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- Chelsea Li, MS
- Rachelle Ambroise, MS



National Heart Lung and Blood Institute

American Heart Association®

Backup | The exact 21-gene spatial outcome

Measured targets

Rxrg
Ppargc1a
Ppargc1b
Cebpa
Cebpb
Esrra
Esrrb

Measured targets

Esrrg
Mitf
Myh6
Tnni3
Myl2
Kcnq1
Jph1

Measured targets

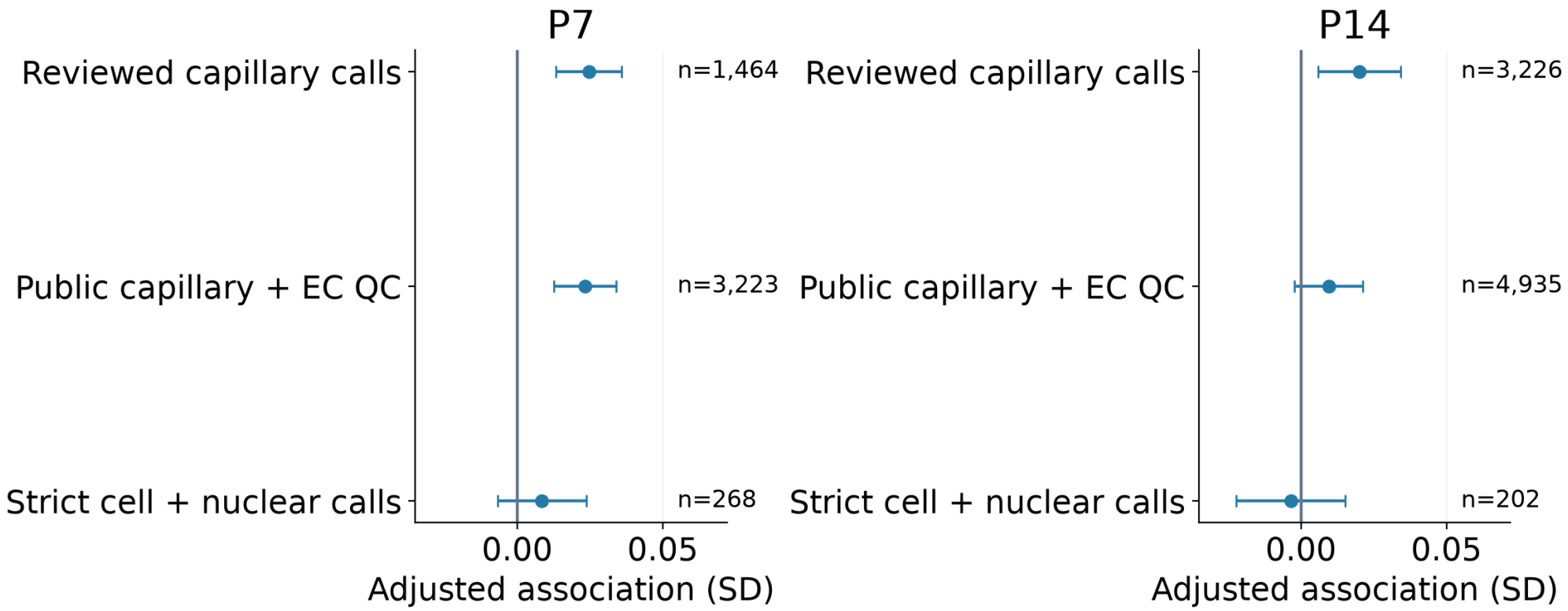
Jph2
Bin1
Lpin1
Pln
Ppara
Ppard
Cpt1b

Module score = mean expression of 21 genes − expression-matched background

Reconstruction: 8 bins, 20 controls/gene, 50 draws; fixed cross-stage reference within each RNA view.

Reviewed 21-gene follow-up (frozen analysis); measured_21gene_panel.tsv; analysis_settings.json

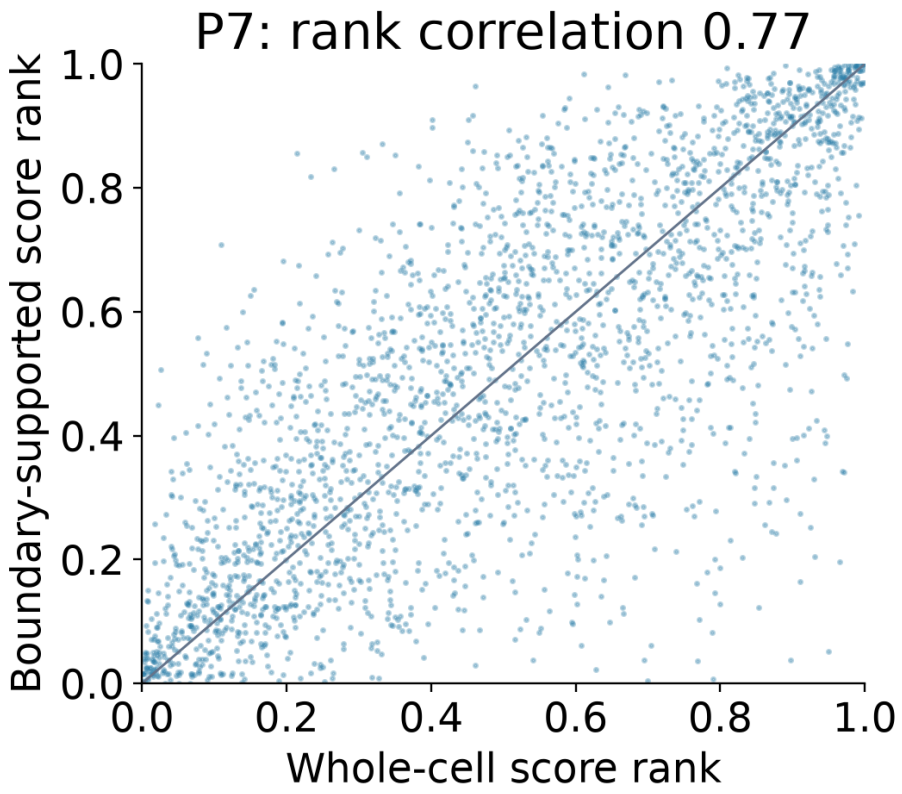
Backup | Capillary labels limit the spatial conclusion

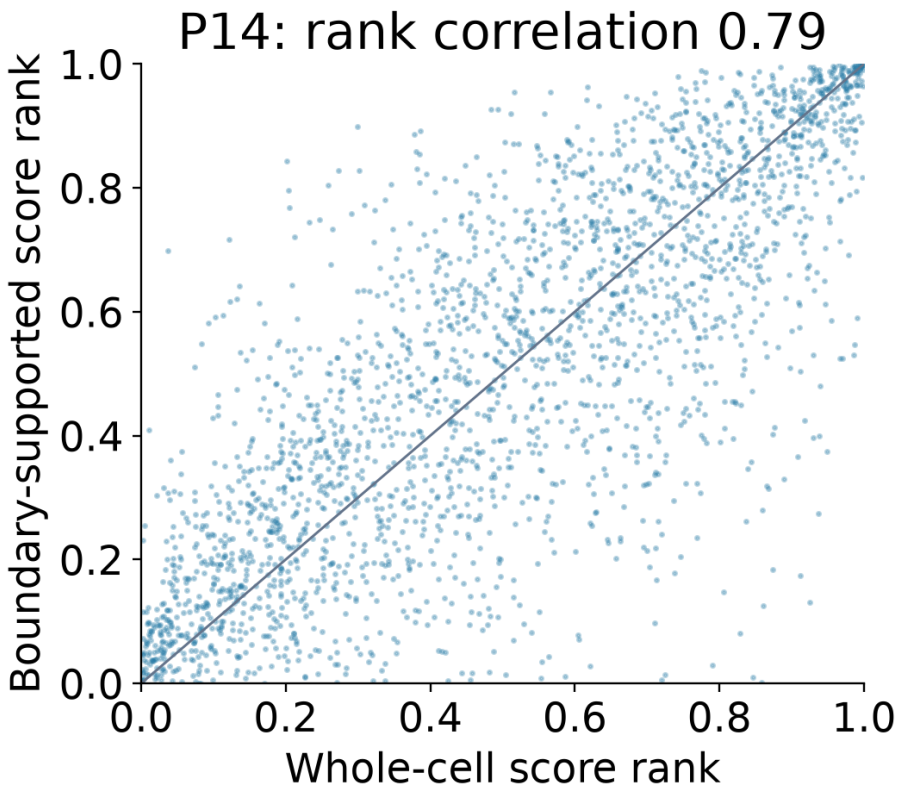


The strict source sets are sparse; neither annotation layer is established biological ground truth.

Reviewed 21-gene follow-up (frozen analysis); source_label_sensitivity.tsv

Backup | Segmentation affects transcript ownership and geometry





Keep valid nucleus-free CM profiles.

Core counts are not nuclear-only truth.

Backup | Fibroblast state change is not lineage tracing

Observed

Different expression families

at different ages

Growing tissue geometry

Possible explanations

State switching

Differential growth / survival

Sampling / label changes

Migration

Discriminating evidence

Timed lineage tracing

Intermediate ages

Registered anatomy

Coherent marker panels

Backup | Communication models organize hypotheses

CytoSPACE

RNA-to-spatial alignment

COMMOT

Spatially compatible communication proposals

SCENIC / SigNET

Candidate regulatory links

Validation

Spatial / animal holdouts

Cell-specific intervention

Backup | Screen assignment and target validation are different

37 / 43

Detectable target probes decreased ≥50%

Probe-signal reduction is an editing proxy, not a measured biallelic knockout percentage.

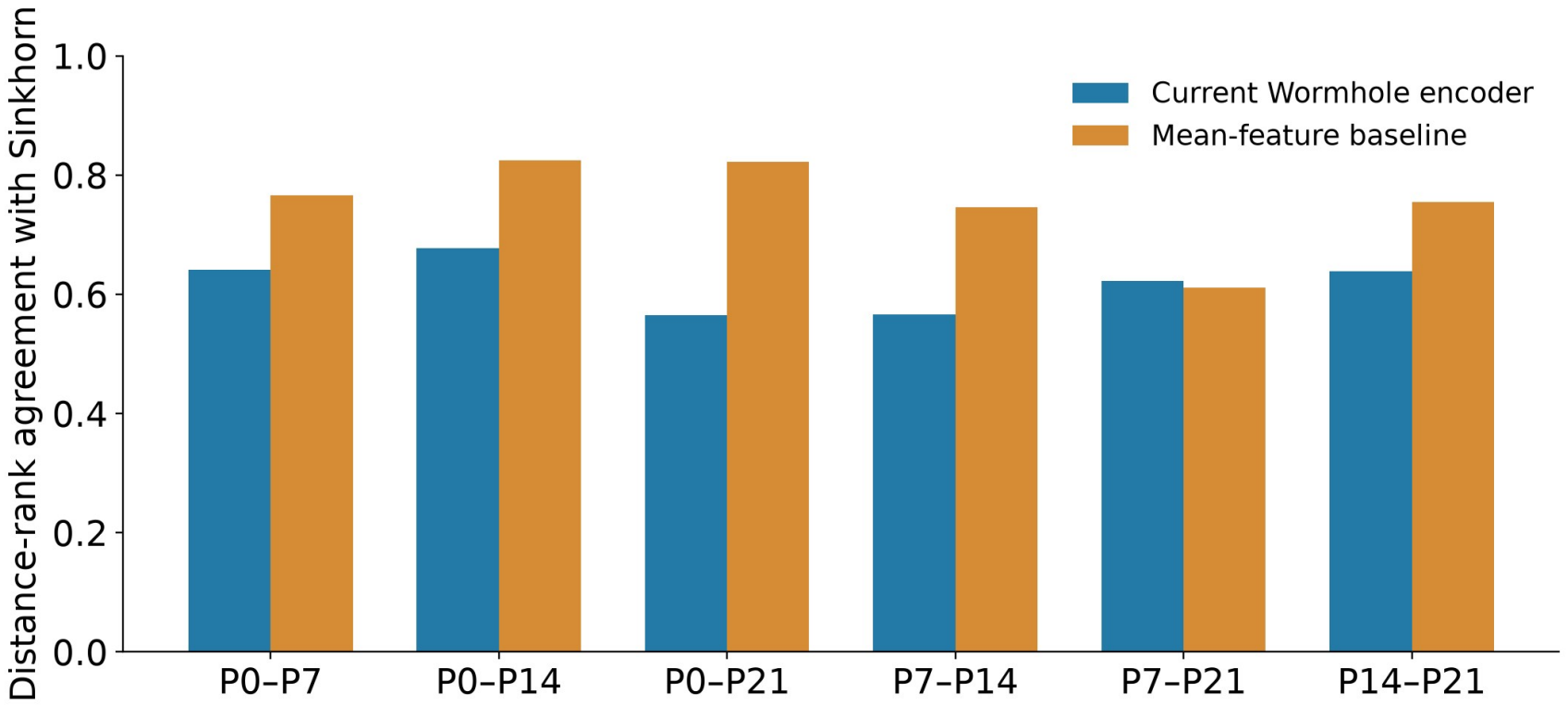
What the readout supports

Target-site RNA evidence

Guide detection ≠ every allele edited

Independent target and phenotype validation remain essential

Backup | Transport can nominate developmental correspondence



Distributional correspondence is not fate tracing; the current encoder does not establish dynamic cardiac niches.

cardiac_developmental_synthesis/cross_stage_metric_assessment.tsv; prior held-out Wormhole assessment

Selected references and source material

Heart atlas / functional genomics

Wang, Dong, Song et al. Spatially guided in vivo single-cell functional genomics of postnatal heart. Nature Cardiovascular Research (2026). DOI: 10.1038/s44161-026-00861-z.

Perinatal coronary expansion

Lu et al. Perinatal angiogenesis from pre-existing coronary vessels via DLL4–NOTCH1 signalling. Nature Cell Biology 23, 967–977 (2021). DOI: 10.1038/s41556-021-00747-1.

CM maturation and regeneration

Guo & Pu. Circulation Research 126, 1086–1106 (2020). Tzahor & Poss. Science 356, 1035–1039 (2017).

Fibroblast developmental context

Hortells et al. A specialized population of Periostin-expressing cardiac fibroblasts contributes to postnatal cardiomyocyte maturation and innervation. PNAS (2020). DOI: 10.1073/pnas.2009119117.

Supplied manuscript panels are credited; exploratory follow-up is kept distinct from published experiments.